

ITC C103
WW2 - M3U1 Midterms Assignment 7

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Answers and Questions:

1. When building a new desktop computer or upgrading an old one, what is the single most important component that determines the range of parts you can use for upgrades in the future?

Answer: Motherboard - It is the main circuit board that connects all the components. Its CPU socket type, RAM type, number of expansion slots, and supported storage connections decide which parts you can install. That is why it sets the limits of any future upgrade.

2. You need to choose a new device to store all your school projects, photos, and personal documents long-term. What type of computer component provides persistent storage, and what are two common examples of it?

Answer: Storage device - This component keeps your files even when the computer is turned off, so it is good for long-term storage of school projects, photos, and documents. Two common examples are the hard disk drive (HDD), which uses spinning magnetic platters and offers large capacity at a lower price, and the solid-state drive (SSD), which has no moving parts and is faster and more durable.

3. A friend wants to know how fast their new computer's CPU is. How is a CPU's speed measured, and what is the function of a heatsink in keeping the CPU from overheating?

Answer: CPU speed is measured in gigahertz (GHz), which shows how many cycles it can perform per second. A higher GHz generally means faster processing, though core count and architecture also matter. The heatsink is a metal component, usually aluminum or copper, that sits on top of the CPU. It absorbs the heat the CPU produces and spreads it into the surrounding air, often with the help of a fan. This keeps the CPU from overheating, slowing down, or getting damaged.

4. You want to add a powerful graphics card to your desktop computer for gaming. What type of expansion slot should you look for on the motherboard to ensure the fastest connection?

Answer: PCIe x16 slot (PCI Express x16). This is the longest PCIe slot on the motherboard and is made for graphics cards. It provides the highest bandwidth, so the GPU can send data quickly to the rest of the system, which matters for gaming performance.

5. You need to change a core setting on your computer, such as the boot order or system clock, before the operating system even loads. What is the name of the program you would access to do this, and how do you typically get to it when you first turn on the computer?

Answer: BIOS/UEFI setup utility. This firmware is stored on the motherboard and runs before the operating system loads. It lets you change core settings like the boot order and system date and time. To open it, press a specific key right after turning on the computer, before Windows starts. The key depends on the manufacturer but is usually Del, F2, F10, F12, or Esc.